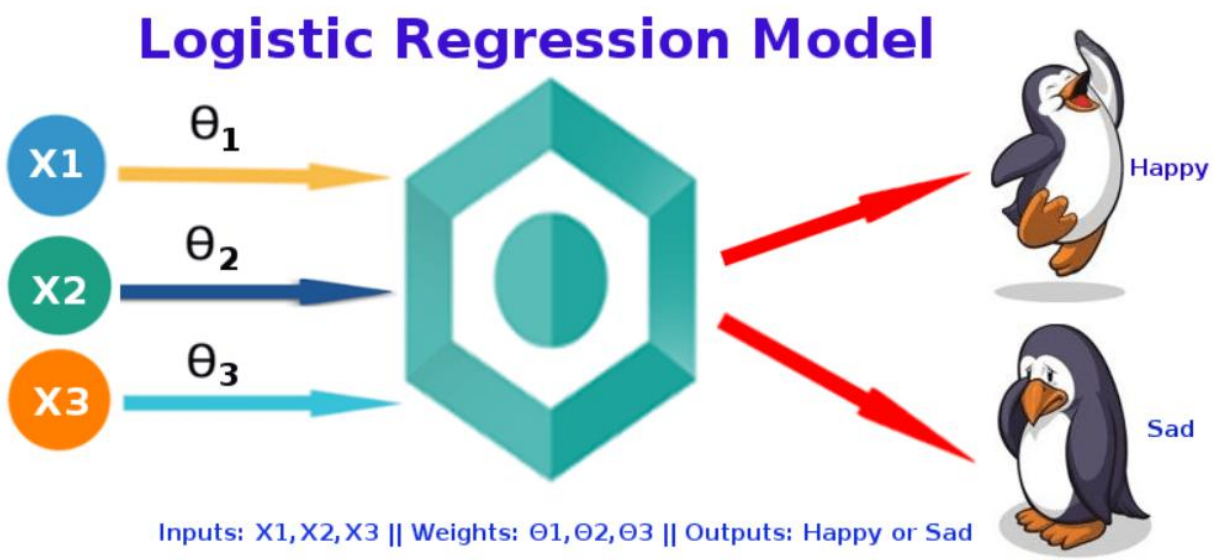
# Logistic Regression

### What is Logistic Regression?

Logistic regression is used for binary classification where we use sigmoid function, that takes input as independent variables and produces a probability value between 0 and 1.

For example, we have two classes Class 0 and Class 1 if the value of the logistic function for an input is greater than 0.5 (threshold value) then it belongs to Class 1 it belongs to Class 0. It's referred to as regression because it is the extension of linear regression but is mainly used for classification problems.




Why do we use Logistic Regression rather than Linear Regression?

If you have this doubt, then you're in the right place, my friend. After reading the definition of logistic regression we now know that it is only used when our dependent variable is binary and in linear regression this dependent variable is continuous.

The second problem is that if we add an outlier in our dataset, the best fit line in linear regression shifts to fit that point.

Now, if we use linear regression to find the best fit line which aims at minimizing the distance between the predicted value and actual value, the line will be like this:

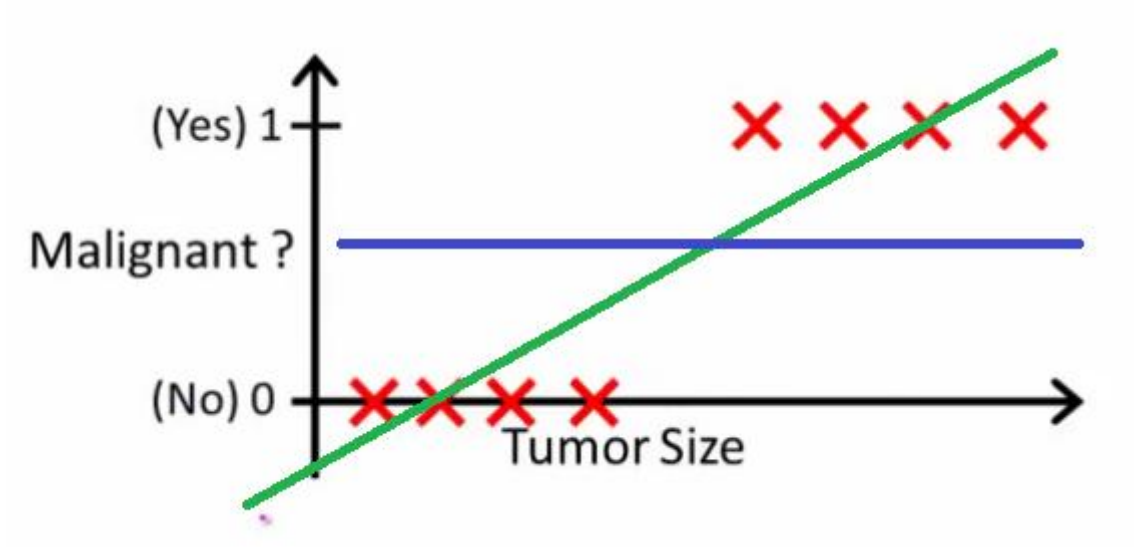



**Image Source:towardsdatascience.com**

Here the threshold value is 0.5, which means if the value of h(x) is greater than 0.5 then we predict malignant tumor (1) and if it is less than 0.5 then we predict benign tumor (0). Everything seems okay here but now let's change it a bit, we add some outliers in our dataset, now this best fit line will shift to that point. Hence the line will be somewhat like this:

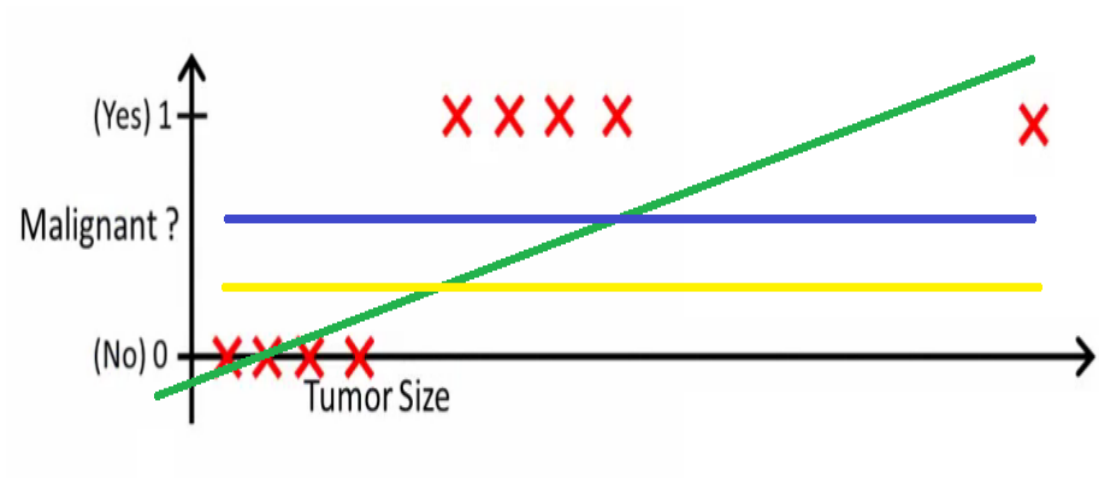



**Image Source:towardsdatascience.com**

Do you see any problem here? The blue line represents the old threshold and the yellow line represents the new threshold which is maybe 0.2 here. To keep our predictions right we had to lower our threshold value. Hence we can say that linear

regression is prone to outliers. Now here if h(x) is greater than 0.2 then only this regression will give correct outputs.

Another problem with linear regression is that the predicted values may be out of range. We know that probability can be between 0 and 1, but if we use linear regression this probability may exceed 1 or go below 0.

To overcome these problems we use Logistic Regression, which converts this straight best fit line in linear regression to an S-curve using the sigmoid function, which will always give values between 0 and 1. How does this work and what's the math behind this will be covered in a later section?

**Key Points:**

- Logistic regression predicts the output of a categorical dependent variable. Therefore, the outcome must be a categorical or discrete value.
- It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1.
- In Logistic regression, instead of fitting a regression line, we fit an "S" shaped logistic function, which predicts two maximum values (0 or 1).

**Logistic Function – Sigmoid Function**

- The sigmoid function is a mathematical function used to map the predicted values to probabilities.
- It maps any real value into another value within a range of 0 and 1. The value of the logistic regression must be between 0 and 1, which cannot go beyond this limit, so it forms a curve like the "S" form.
- The S-form curve is called the Sigmoid function or the logistic function.
- In logistic regression, we use the concept of the threshold value, which defines the probability of either 0 or 1. Such as values above the threshold value tends to 1, and a value below the threshold values tends to 0.

**Types of Logistic Regression**

On the basis of the categories, Logistic Regression can be classified into three types:

1. **Binomial:** In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.
2. **Multinomial:** In multinomial Logistic regression, there can be 3 or more possible unordered types of the dependent variable, such as “cat”, “dogs”, or “sheep”
3. **Ordinal:** In ordinal Logistic regression, there can be 3 or more possible ordered types of dependent variables, such as “low”, “Medium”, or “High”.

**Assumptions of Logistic Regression**

We will explore the assumptions of logistic regression as understanding these assumptions is important to ensure that we are using appropriate application of the model. The assumption include:

1. Independent observations: Each observation is independent of the other. meaning there is no correlation between any input variables.
2. Binary dependent variables: It takes the assumption that the dependent variable must be binary or dichotomous, meaning it can take only two values. For more than two categories SoftMax functions are used.
3. Linearity relationship between independent variables and log odds: The relationship between the independent variables and the log odds of the dependent variable should be linear.
4. No outliers: There should be no outliers in the dataset.
5. Large sample size: The sample size is sufficiently large

**Terminologies involved in Logistic Regression**

Here are some common terms involved in logistic regression:

- **Independent variables:** The input characteristics or predictor factors applied to the dependent variable’s predictions.
- **Dependent variable:** The target variable in a logistic regression model, which we are trying to predict.
- **Logistic function:** The formula used to represent how the independent and dependent variables relate to one another. The logistic function transforms the input variables into a probability value between 0 and 1, which represents the likelihood of the dependent variable being 1 or 0.

- **Odds:** It is the ratio of something occurring to something not occurring. it is different from probability as the probability is the ratio of something occurring to everything that could possibly occur.
- **Log-odds:** The log-odds, also known as the logit function, is the natural logarithm of the odds. In logistic regression, the log odds of the dependent variable are modeled as a linear combination of the independent variables and the intercept.
- **Coefficient:** The logistic regression model's estimated parameters, show how the independent and dependent variables relate to one another.
- **Intercept:** A constant term in the logistic regression model, which represents the log odds when all independent variables are equal to zero.
- **Maximum likelihood estimation:** The method used to estimate the coefficients of the logistic regression model, which maximizes the likelihood of observing the data given the model.

Logistic Function

You must be wondering how logistic regression squeezes the output of linear regression between 0 and 1. If you haven't read my article on Linear Regression then please have a look at it for a better understanding.

Well, there's a little bit of math included behind this and it is pretty interesting trust me.

Let's start by mentioning the formula of logistic function:

$$P = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

How similar it is too linear regression? If you haven't read my article on Linear Regression, then please have a look at it for a better understanding.

*Best Fit Equation in Linear Regression*

We all know the equation of the best fit line in linear regression is:

$$y = \beta_0 + \beta_1 x$$

Let's say instead of y we are taking probabilities (P). But there is an issue here, the value of (P) will exceed 1 or go below 0 and we know that range of Probability is (0-1). To overcome this issue we take ***"odds"*** of P:

$$P = \beta_0 + \beta_1 x$$

$$\frac{P}{1-P} = \beta_0 + \beta_1 x$$

Do you think we are done here? No, we are not. We know that odds can always be positive which means the range will always be (0,+∞ ). Odds are nothing but the ratio of the probability of success and probability of failure. Now the question comes out of so many other options to transform this why did we only take ***'odds'***? Because odds are probably the easiest way to do this, that's it.

The problem here is that the range is restricted and we don't want a restricted range because if we do so then our correlation will decrease. By restricting the range we are actually decreasing the number of data points and of course, if we decrease our data points, our correlation will decrease. It is difficult to model a variable that has a restricted range. To control this we take the ***log of odds*** which has a range from (-∞,+∞).

$$\log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x$$

If you understood what I did here then you have done 80% of the maths. Now we just want a function of P because we want to predict probability right? not log of odds. To do so we will multiply by ***exponent*** on both sides and then solve for P.

$$\exp[\log(\frac{p}{1-p})] \;=\; \exp(\beta_0 + \beta_1 x)$$

$$e^{\ln[\frac{p}{1-p}]} = e^{(\beta_0 + \beta_1 x)}$$

$$\frac{p}{1-p} \;=\; e^{\left(\beta_0 + \beta_1 x\right)}$$

$$p \;= e^{\left(\beta_0 + \beta_1 x\right)} - pe^{\left(\beta_0 + \beta_1 x\right)}$$

$$p \;=\; p[\frac{e^{\left(\beta_0 + \beta_1 x\right)}}{p} - e^{\left(\beta_0 + \beta_1 x\right)}]$$

$$1 = \frac{e^{\left(\beta_0 + \beta_1 x\right)}}{p} - e^{\left(\beta_0 + \beta_1 x\right)}$$

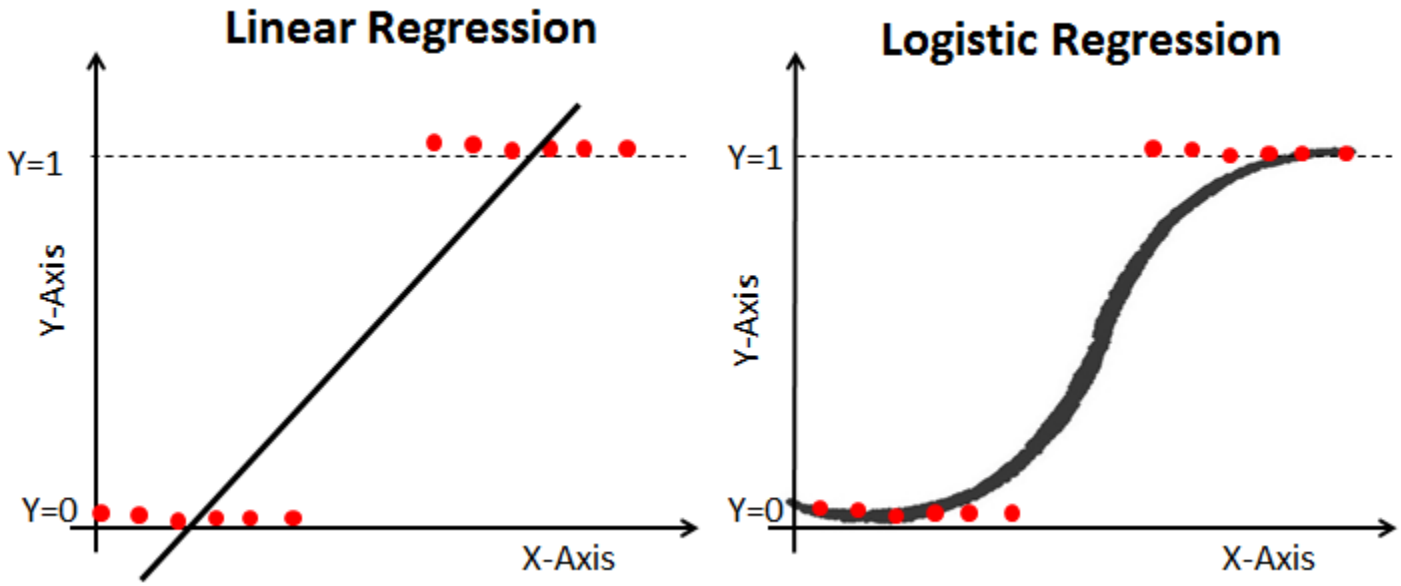
$$p[1 + e^{\left(\beta_0 + \beta_1 x\right)}] \;= e^{\left(\beta_0 + \beta_1 x\right)}$$

$$p \;= \frac{e^{\left(\beta_0 + \beta_1 x\right)}}{1 + e^{\left(\beta_0 + \beta_1 x\right)}}$$

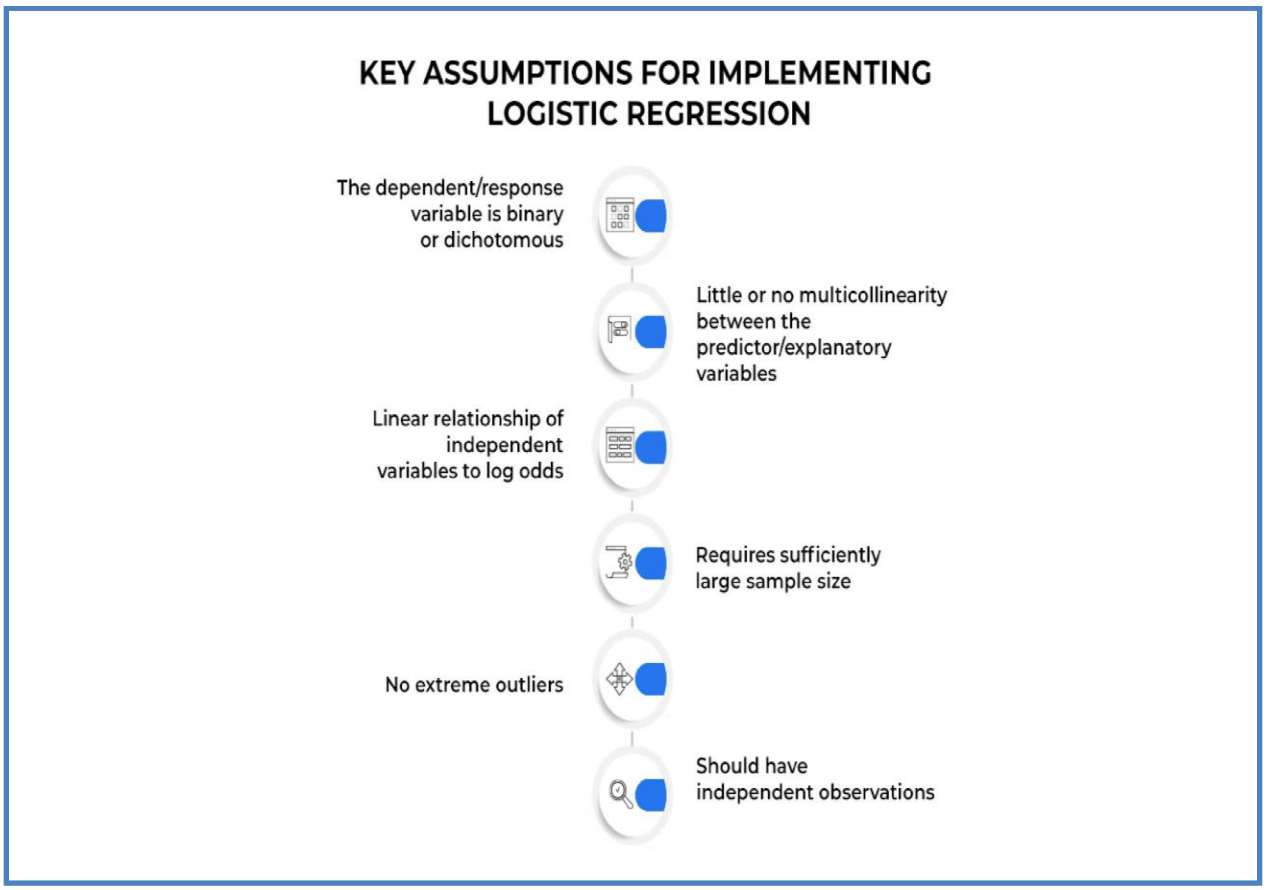
$$Now\ dividing\ by\ e^{\left(\beta_0 + \beta_1 x\right)},\ we\ will\ get$$

$$p \;=\; \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}} \quad This\ is\ our\ sigmoid\ function.$$

Now we have our logistic function, also called a sigmoid function. The graph of a sigmoid function is as shown below. It squeezes a straight line into an S-curve.




Key properties of the logistic regression equation

- **Sigmoid Function**: The logistic regression model, when explained, uses a special "S" shaped curve to predict probabilities. It ensures that the predicted probabilities stay between 0 and 1, which makes sense for probabilities.
- **Straightforward Relationship**: Even though the logistic regression model might seem complex, the relationship between our inputs (like age, height, etc.) and the outcome (like yes/no) is pretty simple to understand. It's like drawing a straight line, but with a curve instead.
- **Coefficients**: These are just numbers that tell us how much each input affects the outcome in the logistic regression model. For example, if age is a predictor, the coefficient tells us how much the outcome changes for every one year increase in age.
- **Best Guess**: We figure out the best coefficients for the logistic regression model by looking at the data we have and tweaking them until our predictions match the real outcomes as closely as possible.

- **Basic Assumptions**: In logistic regression explained, we assume that our observations are independent, meaning one doesn't affect the other. We also assume that there's not too much overlap between our predictors (like age and height), and the relationship between our predictors and the outcome is kind of like a straight line.
- **Probabilities, Not Certainties**: Instead of saying "yes" or "no" directly, logistic regression gives us probabilities, like saying there's a 70% chance it's a "yes" in the logistic regression model. We can then decide on a cutoff point to make our final decision.
- **Checking Our Work**: In logistic regression explained, we have some tools to make sure our predictions are good, like accuracy, precision, recall, and a curve called the ROC curve. These help us see how well our logistic regression model is doing its job.
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